UNIT 1: READING, WRITING, COMPARING AND CALCULATING WHOLE NUMBERS BEYOND 1,000,000

1. PLACE VALUE AND VALUE OF DIGITS

Place value is the position occupied by a digit in a number.

From right to left:
Ones → Tens → Hundreds → Thousands → Ten Thousands
→ Hundred Thousands → Millions

Example:
8,356,421

8 → Millions
3 → Hundred Thousands
5 → Ten Thousands
6 → Thousands
4 → Hundreds
2 → Tens
1 → Ones

VALUE OF A DIGIT

Value = Digit × Place Value

Example:
In 4,835,634:

Place value of 8 = Hundred Thousands
Value of 8 = 8 × 100,000 = 800,000

2. READING AND WRITING LARGE NUMBERS

Large numbers are divided into groups called PERIODS.

Millions | Thousands | Units

Example:
342,602,631

342 | 602 | 631

342 → Millions
602 → Thousands
631 → Units

In words:
Three hundred forty-two million, six hundred two thousand,
six hundred thirty-one.

3. EXPANDED FORM

Expanded form shows the value of every digit.

Example:

24,567,321
= 20,000,000 + 4,000,000 + 500,000 + 60,000
+ 7,000 + 300 + 20 + 1

4. COMPARING LARGE NUMBERS

We use:

> Greater than
< Less than
= Equal to

Rules:

- A number with more digits is greater.
- If they have the same number of digits, compare from left to right.

Example:

6,312,542 > 6,312,452

5. ORDERING NUMBERS

Ascending order:
Smallest → Largest

Descending order:
Largest → Smallest

6. ADDITION OF LARGE NUMBERS

Addition means putting quantities together.

Important words:
- Sum
- Total
- Altogether
- Combined

Steps:
1. Arrange digits according to place value.
2. Start from the ones.
3. Add from right to left.
4. Carry where necessary.

Example:

```
  2,034,315
+ 3,432,541
------------
  5,466,856
```

7. SUBTRACTION OF LARGE NUMBERS

Subtraction means taking one quantity away from another.

The answer is called the DIFFERENCE.

Steps:
1. Arrange numbers according to place value.
2. Start from the right.
3. Subtract each column.
4. Borrow/regroup when necessary.

Example:

```
  6,345,625
- 2,124,304
------------
  4,221,321
```

8. MULTIPLICATION

Multiplication is repeated addition.

Important words:

- Product
- Times
- Each
- Every
- Double
- Triple

For a 2- or 3-digit multiplier:
1. Multiply by the ones.
2. Multiply by the tens.
3. Multiply by the hundreds if present.
4. Add the partial products.

9. DIVISION

Division means sharing equally or finding how many times one number is contained in another.

Example:

20 ÷ 5 = 4

20 = Dividend
5 = Divisor
4 = Quotient

Long division follows:

DIVIDE → MULTIPLY → SUBTRACT → BRING DOWN → REPEAT

10. ROUNDING OFF NUMBERS

General rule:

0, 1, 2, 3, 4 → ROUND DOWN
5, 6, 7, 8, 9 → ROUND UP

To round to:	Look at:
Nearest 10	→ Ones
Nearest 100	→ Tens
Nearest 1,000	→ Hundreds
Nearest 10,000	→ Thousands
Nearest 100,000	→ Ten Thousands
Nearest 1,000,000	→ Hundred Thousands

Examples:

2,458,548 to nearest 10
= 2,458,550

1,257,654 to nearest 1,000
= 1,258,000

7,398,500 to nearest 1,000,000
= 7,000,000

QUICK REVISION

Place value = position of a digit
Value = amount represented by a digit
Sum = answer from addition
Difference = answer from subtraction
Product = answer from multiplication
Quotient = answer from division

Ascending = Smallest → Largest
Descending = Largest → Smallest

Rounding:
0–4 → Down
5–9 → Up

UNIT 2: MULTIPLICATION AND DIVISION OF INTEGERS

1. INTEGERS

An integer is a whole number that can be positive, negative or zero.

Examples:
..., -5, -4, -3, -2, -1, 0, +1, +2, +3, +4, +5, ...

Positive integers → greater than zero
Example: +2, +5, +10

Negative integers → less than zero
Example: -2, -5, -10

Zero (0) is neither positive nor negative.

2. USES OF INTEGERS

Integers can represent:

- Profit and loss
- Rise and fall in temperature
- Positions above and below sea level
- Forward and backward movement
- Gains and losses of points

Examples:

Profit of 10,000 Frw → +10,000 Frw
Loss of 5,000 Frw → -5,000 Frw

5°C rise → +5°C
7°C fall → -7°C

1,000 m above sea level → +1,000 m
200 m below sea level → -200 m

3. MULTIPLICATION OF INTEGERS

Multiplication can be understood as repeated addition.

Example:

$(+3) \times (+2)$

$= (+2) + (+2) + (+2)$
$= +6$

RULES OF SIGNS

$(+) \times (+) = +$
$(+) \times (-) = -$
$(-) \times (+) = -$
$(-) \times (-) = +$

EASY RULE:

SAME SIGNS → POSITIVE
DIFFERENT SIGNS → NEGATIVE

Examples:

(+5) × (+8) = +40
(-7) × (-8) = +56
(+7) × (-9) = -63
(-3) × (+12) = -36

4. MULTIPLICATION USING A NUMBER LINE

On a number line:

Right → Positive direction
Left → Negative direction

Example:

(+3) × (+2)

Start at 0 and make 3 jumps of 2:

0 → 2 → 4 → 6

Therefore:

(+3) × (+2) = +6

5. MULTIPLYING WITHOUT A NUMBER LINE

Step 1: Ignore the signs.
Step 2: Multiply the numbers normally.
Step 3: Apply the sign rule.

Example:

(-5) × (+8)

5 × 8 = 40

Signs are different.

Therefore:

(-5) × (+8) = -40

6. DIVISION OF INTEGERS

Division can be understood as repeated subtraction or equal sharing.

Example:

6 ÷ 2 = 3

RULES OF SIGNS IN DIVISION

(+) ÷ (+) = +
(+) ÷ (-) = -
(-) ÷ (+) = -
(-) ÷ (-) = +

Therefore:

SAME SIGNS → POSITIVE
DIFFERENT SIGNS → NEGATIVE

Examples:

(+18) ÷ (+3) = +6
(-16) ÷ (-4) = +4
(-21) ÷ (+7) = -3
(+12) ÷ (-3) = -4

7. DIVIDING WITHOUT A NUMBER LINE

Step 1: Ignore the signs.
Step 2: Divide normally.
Step 3: Apply the sign rule.

Example:

(-48) ÷ (-8)

48 ÷ 8 = 6

Same signs → Positive

Therefore:

(-48) ÷ (-8) = +6

8. REAL-LIFE PROBLEMS

Example 1: Temperature

Temperature rises by +4°C every hour for 6 hours.

(+4) × (+6) = +24°C

Example 2: Loss

Loss on one shirt = -500 Frw

For 5 shirts:

(-500) × (+5) = -2,500 Frw

Example 3: Football Points

A team loses 3 points in each of 5 games.

(-3) × (+5) = -15 points

QUICK REVISION

Integers = Positive numbers + Negative numbers + Zero

Multiplication → Repeated addition
Division → Repeated subtraction/equal sharing

VERY IMPORTANT RULE:

SAME SIGNS → POSITIVE (+)

(+) × (+) = +
(-) × (-) = +

(+) ÷ (+) = +
(-) ÷ (-) = +

DIFFERENT SIGNS → NEGATIVE (-)

(+) × (-) = -
(-) × (+) = -

(+) ÷ (-) = -
(-) ÷ (+) = -

REMEMBER:

SAME = +
DIFFERENT = -

UNIT 3: POWERS, INDICES, LCM AND GCF

1. INDICES

Indices are used to write repeated multiplication in a shorter form.

Example:
$2^3 = 2 \times 2 \times 2 = 8$

In 2^3:
2 = Base
3 = Exponent (Index)

Example:
$3 \times 3 \times 3 \times 3 = 3^4$

2. MULTIPLICATION OF INDICES

When multiplying powers with the SAME BASE:
Keep the base and ADD the exponents.

Rule:
$a^m \times a^n = a^{m+n}$

Example:
$7^2 \times 7^2$
$= 7^{2+2}$
$= 7^4$

Another example:
$3^3 \times 3^2 \times 3^3$
$= 3^{3+2+3}$
$= 3^8$

3. DIVISION OF INDICES

When dividing powers with the SAME BASE:
Keep the base and SUBTRACT the exponents.

Rule:
$a^m \div a^n = a^{m-n}$

Example:
$8^5 \div 8^3$
$= 8^{5-3}$
$= 8^2$
$= 64$

4. MULTIPLICATION AND DIVISION TOGETHER

Remember:

Multiplication → ADD exponents
Division → SUBTRACT exponents

Example:
$2^6 \times 2^8 \div 2^4$
$= 2^{6+8-4}$
$= 2^{10}$

IMPORTANT:
For any non-zero number:

$a^0 = 1$

Example:
$5^0 = 1$

5. FINDING A MISSING EXPONENT

Example using multiplication:

$4? \times 4^3 = 4^9$

$? + 3 = 9$
$? = 6$

Therefore:
$4^6 \times 4^3 = 4^9$

Example using division:

$4^9 \div 4? = 4^3$

9 - ? = 3
? = 6

Therefore:
$4^9 \div 4^6 = 4^3$

6. LOWEST COMMON MULTIPLE (LCM)

LCM is the SMALLEST number that is a multiple of two or more numbers.

Methods:
• Listing multiples
• Prime factorisation

Example:
Find LCM of 6 and 9.

Multiples of 6:
6, 12, 18, 24, 30, ...

Multiples of 9:
9, 18, 27, 36, ...

First common multiple = 18

Therefore:
LCM(6, 9) = 18

7. USES OF LCM

LCM is used when repeated events need to happen together again.

Example:

One medicine is taken every 12 hours.
Another is taken every 8 hours.

LCM of 12 and 8 = 24

Therefore, they will be taken together again after 24 hours.

8. GREATEST COMMON FACTOR (GCF)

GCF is the GREATEST factor that divides two or more numbers exactly.

Methods:
- Listing factors
- Prime factorisation

Example:
Find GCF of 20 and 24.

Factors of 20:
1, 2, 4, 5, 10, 20

Factors of 24:
1, 2, 3, 4, 6, 8, 12, 24

Common factors:
1, 2, 4

Greatest = 4

Therefore:
GCF(20, 24) = 4

9. USES OF GCF

GCF is useful for:

- Dividing quantities into equal groups
- Cutting objects into the largest equal pieces
- Packing quantities without a remainder

10. RELATIONSHIP BETWEEN LCM AND GCF

For two numbers:

Number 1 × Number 2 = LCM × GCF

To find a missing number:

Missing number =
(LCM × GCF) ÷ Known number

Example:

LCM = 30
GCF = 5
Known number = 10

Missing number
= (30 × 5) ÷ 10
= 15

QUICK REVISION

Index → Short form of repeated multiplication

In 5^3:
5 = Base
3 = Exponent

INDICES:
Multiplication → ADD exponents
$a^m \times a^n = a^{m+n}$

Division → SUBTRACT exponents
$a^m \div a^n = a^{m-n}$

Zero exponent:
$a^0 = 1$

LCM → Smallest common multiple
GCF → Greatest common factor

LCM → Used for repeated events happening together.
GCF → Used for making the largest equal groups.

IMPORTANT FORMULA:

Number 1 × Number 2 = LCM × GCF

UNIT 4: OPERATIONS ON FRACTIONS

1. FRACTIONS

A fraction represents part of a whole.

Examples:

1/2 = one part out of two equal parts
3/4 = three parts out of four equal parts

Fractions are used in:

- Sharing
- Cooking
- Budgeting
- Measurement
- Time
- Area calculations

2. MULTIPLYING A WHOLE NUMBER BY A FRACTION

Steps:

1. Write the whole number over 1.
2. Multiply the numerators.
3. Multiply the denominators.
4. Simplify.

Example:

15 × 2/3

= 15/1 × 2/3
= 30/3
= 10

3. MULTIPLYING A FRACTION BY A WHOLE NUMBER

Use the same method.

Example:

5/12 × 24

= 5/12 × 24/1
= 10

IMPORTANT:

"OF" usually means MULTIPLICATION.

Example:

3/4 of 20

= 3/4 × 20
= 15

4. MULTIPLYING TWO FRACTIONS

Rule:

$a/b \times c/d = ac/bd$

Multiply:

Numerator × Numerator
Denominator × Denominator

Then simplify.

Example:

5/6 × 3/4

= 15/24
= 5/8

5. RECIPROCAL

The reciprocal is found by turning a fraction upside down.

Rule:

Reciprocal of $a/b = b/a$

Examples:

Reciprocal of 2/5 = 5/2

Reciprocal of 3/4 = 4/3

6. DIVIDING FRACTIONS

To divide fractions, use:

KEEP → CHANGE → FLIP

KEEP the first fraction.
CHANGE ÷ into ×.
FLIP the second fraction.

Example:

$2/3 \div 4/5$

$= 2/3 \times 5/4$
$= 10/12$
$= 5/6$

7. DIVIDING A WHOLE NUMBER BY A FRACTION

Write the whole number over 1, then use
KEEP → CHANGE → FLIP.

Example:

$6 \div 2/3$

$= 6/1 \times 3/2$
$= 18/2$
$= 9$

8. DIVIDING A FRACTION BY A WHOLE NUMBER

Write the whole number as a fraction over 1.

Example:

$3/4 \div 2$

$= 3/4 \div 2/1$
$= 3/4 \times 1/2$
$= 3/8$

9. MULTIPLICATION AND DIVISION TOGETHER

Apply the correct rules and simplify carefully.

Example:

$1/2 \times 2/3 \div 5/6$

Change division to multiplication:

$= 1/2 \times 2/3 \times 6/5$

Simplify:

$= 2/5$

10. FRACTION WORD PROBLEMS

Steps:

1. Read the question carefully.
2. Identify the given information.
3. Choose the correct operation.
4. Calculate.
5. Simplify.
6. Write the correct unit.

Example:

3/5 of the total number of questions = 30.

Total questions:

$30 \div 3/5$

$= 30 \times 5/3$

= 50 questions.

QUICK REVISION

MULTIPLICATION:

$a/b \times c/d = ac/bd$

"OF" = MULTIPLICATION

RECIPROCAL:

a/b → b/a

DIVISION:

a/b ÷ c/d

= a/b × d/c
REMEMBER:

KEEP → CHANGE → FLIP
Keep the first fraction.
Change ÷ to ×.
Flip the second fraction.

Always simplify the final answer where possible.

UNIT 5: ROUNDING AND CONVERSION OF DECIMAL NUMBERS

1. DECIMAL PLACE VALUES

Example:
23.456789

4 → Tenths
5 → Hundredths
6 → Thousandths
7 → Ten-thousandths
8 → Hundred-thousandths
9 → Millionths

2. ROUNDING DECIMALS

Steps:

1. Identify the place value required.
2. Look at the digit immediately to the right.
3. Apply the rounding rule.

RULE:

0, 1, 2, 3, 4 → ROUND DOWN
5, 6, 7, 8, 9 → ROUND UP

Example:

12.4637 to the nearest hundredth

Hundredths digit = 6
Next digit = 3

Since $3 < 5$:

$12.4637 \approx 12.46$

3. ROUNDING TO DIFFERENT DECIMAL PLACES

Nearest tenth → look at hundredths digit

Nearest hundredth → look at thousandths digit

Nearest thousandth → look at ten-thousandths digit

Nearest ten-thousandth → look at hundred-thousandths digit

Nearest hundred-thousandth → look at millionths digit

4. CONVERTING FRACTIONS TO DECIMALS

Rule:

Numerator ÷ Denominator

Example:

3/4

$= 3 \div 4$
$= 0.75$

Therefore:

$3/4 = 0.75$

5. CONVERTING DECIMALS TO FRACTIONS

Write the decimal over:

10 → 1 decimal place
100 → 2 decimal places
1,000 → 3 decimal places
10,000 → 4 decimal places

Then simplify.

Example:

0.25

= 25/100
= 1/4

Example:

0.236

= 236/1000
= 59/250

6. CONVERTING MIXED DECIMALS TO FRACTIONS

Example:

4.09

= 409/100

or:

4 9/100

7. SOLVING PROBLEMS WITH DECIMALS AND FRACTIONS

Steps:

1. Read the problem carefully.
2. Identify whether the answer should be a fraction or decimal.
3. Do the required calculation.
4. Convert the final answer.

5. Simplify if necessary.

Example:

2.35 m + 1.265 m + 0.75 m

= 4.365 m

Convert to a fraction:

4.365
= 4365/1000
= 873/200
= 4 73/200 m
QUICK REVISION
ROUNDING:
0–4 → Round down
5–9 → Round up
FRACTION → DECIMAL
Numerator ÷ Denominator

Example:

1/4 = 1 ÷ 4 = 0.25

DECIMAL → FRACTION

Use 10, 100, 1,000, 10,000, etc.
depending on the number of decimal places.

Example:

0.65
= 65/100
= 13/20

REMEMBER:

1 decimal place → /10
2 decimal places → /100
3 decimal places → /1000

Always simplify fractions where possible.

UNIT 6: PERCENTAGES, RATIOS AND RELATED PROBLEMS

1. PERCENTAGE

Percentage means "per hundred".

Symbol: %

Example:
25% = 25/100 = 0.25

2. PERCENTAGE → DECIMAL

Divide by 100.

Examples:

75% = 75 ÷ 100 = 0.75

12.5% = 12.5 ÷ 100 = 0.125

3. DECIMAL → PERCENTAGE

Multiply by 100%.

Examples:

0.35 × 100% = 35%

0.8 × 100% = 80%

4. PERCENTAGE → FRACTION

Steps:

1. Remove %.
2. Put the number over 100.
3. Simplify.

Example:

40%

= 40/100
= 2/5

5. FRACTION → PERCENTAGE

Rule:

Fraction × 100%

Example:

3/4 × 100%
= 75%

6. FINDING A PERCENTAGE OF A QUANTITY

Rule:

Percentage of quantity
= Percentage/100 × Quantity

Example:

20% of 500

= 20/100 × 500
= 100

7. FINDING PERCENTAGE

Rule:

Percentage = Part/Whole × 100%

Example:

15 learners out of 60 are absent.

Percentage absent:

15/60 × 100%
= 25%

8. INCREASING A QUANTITY BY A PERCENTAGE

Example:

Increase 10,000 Frw by 20%.

Increase:

$20/100 \times 10{,}000$
$= 2{,}000$ Frw

New amount:

$10{,}000 + 2{,}000$
$= 12{,}000$ Frw

Shortcut:

New amount =
Original × (100 + increase%)/100

9. DECREASING A QUANTITY BY A PERCENTAGE

Example:

Decrease 20,000 Frw by 10%.

Decrease:

$10/100 \times 20{,}000$
$= 2{,}000$ Frw

New amount:

$20{,}000 - 2{,}000$
$= 18{,}000$ Frw

Shortcut:

New amount =
Original × (100 - decrease%)/100

10. FINDING THE ORIGINAL AMOUNT

After an increase:

If a number increases by 15%,
the new amount represents 115%.

Example:

115% = 34,500

1% = 34,500 ÷ 115
= 300

100% = 30,000

Original amount = 30,000

After a decrease:

If an amount decreases by 12%,
the remaining amount represents:

100% - 12% = 88%

11. PERCENTAGE INCREASE AND DECREASE

Percentage Increase:

Increase = New amount - Old amount

Percentage increase
= Increase/Old amount × 100%

Percentage Decrease:

Decrease = Old amount - New amount

Percentage decrease
= Decrease/Old amount × 100%

Example:

600 increases to 840.

Increase = 840 - 600

= 240

Percentage increase:

240/600 × 100%
= 40%

12. PROFIT AND LOSS

CP = Cost Price
SP = Selling Price

PROFIT:

If SP > CP, there is a profit.

Profit = SP - CP

Percentage Profit
= Profit/CP × 100%

LOSS:

If SP < CP, there is a loss.

Loss = CP - SP

Percentage Loss
= Loss/CP × 100%

Example:

CP = 10,000 Frw
SP = 12,500 Frw

Profit = 2,500 Frw

Percentage profit:

2,500/10,000 × 100%
= 25%

13. RATIO

A ratio compares two or more quantities.

Example:

20 boys and 30 girls

Boys : Girls

20 : 30

Simplify:

2 : 3

14. SHARING IN A RATIO

Example:

Share 360 in the ratio 4 : 5.

Total parts:

$4 + 5 = 9$

First share:

$4/9 \times 360 = 160$

Second share:

$5/9 \times 360 = 200$

15. INCREASING OR DECREASING IN A RATIO

Rule:

New quantity
= Old quantity × New part/Old part

QUICK REVISION

Percentage means → Out of 100

Percentage → Decimal:
÷ 100

Decimal → Percentage:
× 100%

Percentage → Fraction:
Put over 100 and simplify.

Fraction → Percentage:
× 100%

IMPORTANT FORMULAS

Percentage
= Part/Whole × 100%

Percentage increase
= Increase/Original × 100%

Percentage decrease
= Decrease/Original × 100%

Profit = SP - CP

Loss = CP - SP

Percentage Profit
= Profit/CP × 100%

Percentage Loss
= Loss/CP × 100%

RATIO:

a : b

To share in a ratio:

1. Add the ratio parts.
2. Divide the total by the number of parts.
3. Multiply by each ratio part.

UNIT 7: RELATIONSHIP BETWEEN VOLUME, CAPACITY AND MASS

1. CAPACITY

Capacity is the amount of liquid a container can hold.

Common units:

kℓ → kilolitre
hℓ → hectolitre
daℓ → decalitre
ℓ → litre
dℓ → decilitre
cℓ → centilitre
mℓ → millilitre

2. MASS

Mass is the quantity of matter in an object.

Common units:

t → tonne
kg → kilogram
hg → hectogram
dag → decagram
g → gram
dg → decigram
cg → centigram
mg → milligram

3. VOLUME

Volume is the space occupied by an object.

Common units:

m^3 → cubic metre
dm^3 → cubic decimetre
cm^3 → cubic centimetre
mm^3 → cubic millimetre

4. VOLUME OF A CUBOID

Formula:

Volume = Length × Width × Height

Example:

Length = 5 m
Width = 3 m
Height = 2 m

Volume:

$5 \times 3 \times 2$
$= 30\ m^3$

5. IMPORTANT CAPACITY CONVERSIONS

1 kℓ = 1,000 ℓ

1 ℓ = 10 dℓ

1 ℓ = 100 cℓ

1 ℓ = 1,000 mℓ

6. IMPORTANT MASS CONVERSIONS

1 tonne = 1,000 kg

1 kg = 1,000 g

1 g = 1,000 mg

7. IMPORTANT VOLUME CONVERSIONS

$1\ m^3 = 1{,}000\ dm^3$

$1\ dm^3 = 1{,}000\ cm^3$

$1\ cm^3 = 1{,}000\ mm^3$

8. RELATIONSHIP BETWEEN WATER, MASS, CAPACITY AND VOLUME

For water:

1 ℓ = 1 kg = 1 dm^3

Also:

1 mℓ = 1 g = 1 cm^3

This means:

1 litre of water has a mass of 1 kg.

1 millilitre of water has a mass of 1 g.

9. CONVERTING WATER BETWEEN UNITS

Example 1:

Convert 13 dm^3 of water to kg.

Since:

1 dm^3 = 1 kg

Therefore:

13 dm^3 = 13 kg

Example 2:

Convert 7,800 ℓ to m^3.

Since:

1,000 ℓ = 1 m^3

7,800 ÷ 1,000
= 7.8

Therefore:

7,800 ℓ = 7.8 m^3

10. REAL-LIFE APPLICATION

A person fetches water using a 20-litre jerrycan
twice a day.

Total water:

20×2
$= 40\ \ell$

Since:

$1\ \ell$ of water = 1 kg

Therefore:

$40\ \ell = 40$ kg

11. WATER TANK EXAMPLE

A tank contains 10,000 litres of water.

Volume:

$1\ \ell = 1\ dm^3$

Therefore:

$10{,}000\ \ell = 10{,}000\ dm^3$

Mass:

$1\ \ell = 1$ kg

Therefore:

$10{,}000\ \ell = 10{,}000$ kg

Since:

1,000 kg = 1 tonne

10,000 kg = 10 tonnes

QUICK REVISION

CAPACITY
= Amount of liquid a container can hold.

MASS
= Quantity of matter in an object.

VOLUME
= Space occupied by an object.

IMPORTANT FORMULA

Volume of cuboid:

$V = L \times W \times H$

IMPORTANT CONVERSIONS

1,000 mℓ = 1 ℓ

1,000 ℓ = 1 kℓ

1,000 g = 1 kg

1,000 kg = 1 tonne

$1\ m^3 = 1{,}000\ dm^3$

$1\ dm^3 = 1{,}000\ cm^3$

$1\ cm^3 = 1{,}000\ mm^3$

VERY IMPORTANT FOR WATER

$1\ ℓ = 1\ kg = 1\ dm^3$

$1\ mℓ = 1\ g = 1\ cm^3$

UNIT 8: SPEED, DISTANCE AND TIME

1. 12-HOUR AND 24-HOUR CLOCK

The 12-hour clock uses:

A.M. → Before midday
P.M. → After midday

The 24-hour clock runs from:
00:00 to 23:59

Examples:

6:45 A.M. = 06:45
3:20 P.M. = 15:20
8:44 P.M. = 20:44

2. CONVERTING TIME

A.M. → 24-hour time:

For most A.M. times, keep the hour.

Example:

6:24 A.M. = 06:24

P.M. → 24-hour time:

Add 12 hours.

Example:

11:28 P.M.

11:28 + 12:00
= 23:28

24-hour → 12-hour:

If above 12:00, subtract 12 hours and use P.M.

Example:

15:54 - 12:00
= 3:54 P.M.

Midnight example:

00:25 = 12:25 A.M.

3. TIME ZONES

A time zone is a region that uses the same standard time.

The world has 24 time zones.

Neighbouring time zones generally differ by 1 hour.

IMPORTANT RULE:

EAST → ADD time
WEST → SUBTRACT time

Example:

If it is 12:00 in the reference zone:

2 zones East:
12:00 + 2 hours = 14:00

7 zones West:
12:00 - 7 hours = 05:00

4. SPEED

Speed tells us how fast or slowly an object moves.

Formula:

SPEED = DISTANCE ÷ TIME

S = D ÷ T

Example:

A motorcyclist travels 210 km in 3 hours.

Speed:

210 ÷ 3
= 70 km/h

5. DISTANCE

Distance tells us how far an object travels.

Formula:

DISTANCE = SPEED × TIME

$D = S \times T$

Example:

Speed = 60 km/h
Time = 4 hours

Distance:

60 × 4
= 240 km

6. TIME

Time tells us how long a journey takes.

Formula:

TIME = DISTANCE ÷ SPEED

$T = D \div S$

Example:

Distance = 180 km
Speed = 60 km/h

Time:

180 ÷ 60
= 3 hours

7. CONVERTING km/h TO m/s

Rule:

km/h → m/s

Multiply by 5/18

Example:

72 km/h

= 72 × 5/18
= 20 m/s

8. CONVERTING m/s TO km/h

Rule:

m/s → km/h

Multiply by 18/5

Example:

20 m/s

= 20 × 18/5
= 72 km/h

9. AVERAGE SPEED

Formula:

AVERAGE SPEED =
TOTAL DISTANCE ÷ TOTAL TIME

Example:

A taxi travels:

160 km + 200 km = 360 km

Total time = 6 hours

Average speed:

360 ÷ 6
= 60 km/h

10. IMPORTANT CONVERSIONS

TIME:

1 hour = 60 minutes

1 minute = 60 seconds

1 hour = 3,600 seconds

DISTANCE:

1 km = 1,000 m

SPEED:

km/h → m/s = × 5/18

m/s → km/h = × 18/5

QUICK REVISION

TIME ZONES:

East → ADD
West → SUBTRACT

FORMULAS:

Speed = Distance ÷ Time
Distance = Speed × Time
Time = Distance ÷ Speed
Average Speed =
Total Distance ÷ Total Time

FORMULA TRIANGLE:

DISTANCE

———————

SPEED | TIME

REMEMBER:
S = D ÷ T
D = S × T
T = D ÷ S

km/h → m/s = × 5/18

m/s → km/h = × 18/5

UNIT 9: SIMPLE INTEREST AND PROBLEMS INVOLVING SAVING

1. IMPORTANT TERMS

Principal (P)
= The amount of money borrowed, saved or lent.

Rate (R)
= The percentage used to calculate interest.

Example:
10% per annum = 10% per year.

Time (T)
= The period for which money is saved, invested or borrowed.

In simple interest, time is usually expressed in years.

Simple Interest (S.I.)
= Extra money earned or paid on the principal.

2. SIMPLE INTEREST FORMULA

S.I. = (P × R × T) / 100

Where:

P = Principal
R = Rate
T = Time
S.I. = Simple Interest

Example:

P = 100,000 Frw
R = 10%
T = 2 years

S.I.

$= (100{,}000 \times 10 \times 2) / 100$

= 20,000 Frw

3. TIME GIVEN IN MONTHS

Since the formula uses years:

12 months = 1 year

Therefore:

Time in years
= Number of months / 12

Example:

6 months

= 6/12 year
= 1/2 year

4. FINDING THE PRINCIPAL

Formula:

$P = (S.I. \times 100) / (R \times T)$

Example:

S.I. = 10,000 Frw
R = 5%
T = 2 years

P

= (10,000 × 100) / (5 × 2)

= 100,000 Frw

5. FINDING THE RATE

Formula:

R = (S.I. × 100) / (P × T)

6. FINDING THE TIME

Formula:

T = (S.I. × 100) / (P × R)

7. TOTAL AMOUNT

Amount is the principal plus the interest.

Formula:

A = P + S.I.

Example:

Principal = 100,000 Frw
Interest = 20,000 Frw

Amount:

100,000 + 20,000
= 120,000 Frw

8. SAVING

Savings is the money remaining after expenditure.

Formula:

SAVINGS = INCOME - EXPENDITURE

Example:

Income = 10,000 Frw

Expenditure:

Books = 3,000 Frw
Graph book = 500 Frw
Handkerchief = 500 Frw
Pens = 1,000 Frw
Mathematics set = 2,000 Frw

Total expenditure:

= 7,000 Frw

Savings:

10,000 - 7,000

= 3,000 Frw

9. WAYS OF SAVING AND INVESTING

Examples include:

• Depositing money in a bank
• Saving through a SACCO
• Buying shares
• Investing in companies
• Investing in cooperative societies

10. IMPORTANCE OF SAVING

Saving can help a person:

• Have money for future needs
• Earn interest
• Invest for future benefits
• Improve future living conditions

QUICK REVISION

P = Principal
R = Rate
T = Time
S.I. = Simple Interest
A = Amount

MAIN FORMULA:

$$S.I. = (P \times R \times T) / 100$$

PRINCIPAL:

$$P = (S.I. \times 100) / (R \times T)$$

RATE:

$$R = (S.I. \times 100) / (P \times T)$$

TIME:

$$T = (S.I. \times 100) / (P \times R)$$

TOTAL AMOUNT:

$$A = P + S.I.$$

SAVINGS:

Savings = Income - Expenditure

REMEMBER:

12 months = 1 year

When time is given in months,
convert it into years before using
the simple interest formula.

UNIT 10: EQUIVALENT EXPRESSIONS AND NUMBER SEQUENCES

1. ALGEBRAIC EXPRESSIONS

An algebraic expression is a mathematical statement that may contain:

• Numbers
• Letters
• Operation signs

Examples:

3x + 5
2a - 4
6m
x + 7

Letters such as x, y, m and n can represent unknown numbers.

2. WRITING EXPRESSIONS FROM WORDS

Some words help us know the operation to use.

Addition words:

• Sum
• Total
• More than
• Added to
• Increased by

Example:

A number x increased by 5

= x + 5

Subtraction words:

• Difference
• Less than

- Reduced by
- Decreased by

Example:

A number y decreased by 3

= y - 3

Multiplication words:

- Product
- Times
- Multiplied by

Example:

5 times x

= 5x

3. EQUIVALENT EXPRESSIONS

Equivalent expressions are different expressions that give the same value.

Example:

2(x + 3)

Expand:

= 2x + 6

Therefore:

2(x + 3) and 2x + 6 are equivalent.

4. EXPANDING BRACKETS

Multiply the number outside the bracket by every term inside.

Example:

3(x + 4)

= 3x + 12

Example:

2(a - 5)

= 2a - 10

5. COMBINING LIKE TERMS

Like terms have the same letters.

Example:

3x + 2x

= 5x

Example:

4a + 3 + 2a + 5

= 6a + 8

6. NUMBER SEQUENCES

A sequence is a set of numbers arranged in a particular order according to a rule.

Example:

2, 5, 8, 11, 14, ...

Rule:

Add 3 each time.

7. TERM

Each number in a sequence is called a term.

Example:

2, 5, 8, 11, ...

1st term = 2
2nd term = 5
3rd term = 8
4th term = 11

8. COMMON DIFFERENCE

The common difference is the fixed amount added or subtracted to get the next term.

Example:

3, 7, 11, 15, ...

Difference:

7 - 3 = 4

Therefore:

Common difference = 4

9. FINDING MISSING TERMS

Example:

1, 5, 9, 13, __, __, __

Common difference = 4

Therefore:

17, 21, 25

10. GENERAL RULE OF A SEQUENCE

The general rule is a formula used to find any term.

For many linear sequences:

General term
= (Common difference × n) + or - a constant

Example:

3, 7, 11, 15, ...

Common difference = 4

Start with:

$4n$

When $n = 1$:

$4(1) = 4$

But first term is 3.

Therefore subtract 1.

General rule:

$4n - 1$

11. Nth TERM

The nth term is the term at position n.

Example:

Sequence:

3, 7, 11, 15, ...

General rule:

$4n - 1$

Find the 8th term:

$4(8) - 1$

$= 32 - 1$

$= 31$

12. GENERATING A SEQUENCE FROM A RULE

Example:

Rule:

$3n + 2$

For $n = 1$:

$3(1) + 2 = 5$

For $n = 2$:

$3(2) + 2 = 8$

For $n = 3$:

$3(3) + 2 = 11$
Sequence:

5, 8, 11, 14, ...
QUICK REVISION
Expression
= Mathematical statement containing numbers, letters or operation signs.
Equivalent expressions
= Different expressions with the same value.
Sequence
= Numbers arranged according to a rule.

Term
= Each number in a sequence.

Common difference
= Fixed amount added or subtracted.

General rule
= Formula used to find any term.

Nth term
= Term at position n.

REMEMBER:

To find the pattern:

1. Check the difference between terms.
2. Identify the common difference.
3. Continue the sequence using the same rule.

For many linear sequences:

General term
= (Common difference × n) + or – constant

UNIT 11: REGULAR POLYGONS AND BEARINGS

1. POLYGONS

A polygon is a 2-dimensional shape formed by connected straight line segments.

A REGULAR POLYGON has:
• All sides equal
• All interior angles equal

Otherwise, it is an irregular polygon.

2. NAMES OF POLYGONS

3 sides → Triangle
4 sides → Quadrilateral
5 sides → Pentagon
6 sides → Hexagon
7 sides → Heptagon
8 sides → Octagon
9 sides → Nonagon
10 sides → Decagon

3. INTERIOR AND EXTERIOR ANGLES

Interior angle = Angle inside a polygon.

Exterior angle = Angle outside a polygon formed when one side is extended.

IMPORTANT:

Interior angle + Exterior angle = 180°

Example:

Interior angle = 108°

Exterior angle:

= 180° - 108°
= 72°

4. SUM OF INTERIOR ANGLES

Formula:

Sum of interior angles = (n - 2) × 180°

Where:
n = number of sides

Example: Hexagon

n = 6

(6 - 2) × 180°
= 4 × 180°
= 720°

5. EXTERIOR ANGLES

The sum of exterior angles of a regular polygon is always:

360°

One exterior angle:

Exterior angle = 360° ÷ n

Example: Regular hexagon

360° ÷ 6
= 60°

6. ONE INTERIOR ANGLE

Formula:

Interior angle = $180° - (360° \div n)$

Example: Regular hexagon

$= 180° - 60°$
$= 120°$

7. FINDING NUMBER OF SIDES

Formula:

Number of sides = $360° \div$ Exterior angle

Example:

Exterior angle = $45°$

Number of sides:

$360° \div 45°$
$= 8$

Therefore, it is an octagon.

8. PERIMETER OF A REGULAR POLYGON

Formula:

Perimeter = Number of sides × Length of one side

Example:

Regular pentagon with side 8 cm:

$P = 5 \times 8$
$= 40$ cm

To find one side:

Side = Perimeter ÷ Number of sides

9. AREA OF A REGULAR POLYGON

Formula:

Area = 1/2 × Perimeter × Apothem

The apothem is the perpendicular distance from the centre of a regular polygon to one of its sides.

To find the apothem:

Apothem = (2 × Area) ÷ Perimeter

10. COMPASS DIRECTIONS

Main directions:

```
       N
       |
   NW  |  NE
       |
W -------+------- E
       |
   SW  |  SE
       |
       S
```

N → North
NE → North-East
E → East
SE → South-East
S → South
SW → South-West
W → West
NW → North-West

11. BEARINGS

A bearing shows direction using an angle.

RULES:

- Start from NORTH.
- Measure CLOCKWISE.
- Write bearings using THREE DIGITS.

Examples:

North → 000°
East → 090°
South → 180°
West → 270°

Example:

045° means 45° clockwise from North.

12. REVERSE BEARING

To find the opposite/reverse bearing:

If bearing is less than 180°:
ADD 180°.

If bearing is 180° or more:
SUBTRACT 180°.

Examples:

045° + 180° = 225°

220° - 180° = 040°

13. TILING

Tiling means covering a surface with tiles to form a pattern or design.

To find the number of tiles:

Step 1:
Find the area of the surface.

Step 2:
Find the area of one tile.

Step 3:

Number of tiles
= Total surface area ÷ Area of one tile

QUICK REVISION

Interior + Exterior = 180°

Sum of interior angles:
= (n - 2) × 180°

Sum of exterior angles:
= 360°

One exterior angle:
= 360° ÷ n

One interior angle:
= 180° - (360° ÷ n)

Number of sides:
= 360° ÷ Exterior angle

Perimeter:
= Number of sides × Side

Side:
= Perimeter ÷ Number of sides

Area:
= 1/2 × Perimeter × Apothem

Apothem:
= (2 × Area) ÷ Perimeter

BEARINGS:

Start → North
Direction → Clockwise
Use → Three digits

Reverse bearing:
Below 180° → Add 180°
180° or above → Subtract 180°

UNIT 12: CONSTRUCTION OF POLYGONS AND NETS FOR CUBOIDS AND PRISMS

1. GEOMETRICAL CONSTRUCTION

Geometrical construction means drawing shapes accurately
using mathematical instruments.

Important instruments:

• Ruler
• Protractor
• Pair of compasses
• Sharp pencil

2. DRAWING TRIANGLES

A triangle can be drawn when sides and/or angles are given.

Example:

AB = 6 cm
AC = 4 cm
Angle BAC = 45°

Steps:

1. Draw AB = 6 cm.
2. At A, construct 45°.
3. Measure AC = 4 cm.
4. Join B to C.

3. DRAWING A SQUARE

A square has:

• 4 equal sides
• 4 right angles
• Each angle = 90°

Steps:

1. Draw one side.
2. Construct 90° angles at both ends.
3. Measure equal sides.
4. Join the remaining points.

4. DRAWING A RECTANGLE

A rectangle has:

- 4 sides
- Opposite sides equal
- Opposite sides parallel
- 4 right angles

Steps:

1. Draw the length.
2. Construct 90° angles.
3. Measure the width.
4. Join the remaining points.

5. CONSTRUCTING A REGULAR POLYGON

A regular polygon has:

- All sides equal
- All angles equal

General method:

1. Find the exterior angle.
2. Find the interior angle.
3. Draw a baseline.
4. Construct the required angle.
5. Measure equal sides.
6. Repeat until the polygon closes.

6. CENTRE ANGLE

Formula:

Centre angle = 360° ÷ Number of sides

The book also states:

Centre angle = Exterior angle

COMMON CENTRE ANGLES

Pentagon:
$360° \div 5 = 72°$

Hexagon:
$360° \div 6 = 60°$

Heptagon:
$360° \div 7 \approx 51°$

Octagon:
$360° \div 8 = 45°$

Nonagon:
$360° \div 9 = 40°$

Decagon:
$360° \div 10 = 36°$

7. CONSTRUCTING REGULAR POLYGONS IN A CIRCLE

General steps:

1. Draw a circle.
2. Mark its centre.
3. Calculate the centre angle.
4. Construct the centre angle.
5. Mark equal points around the circle.
6. Join consecutive points.

Example: Regular Octagon

Number of sides = 8

Centre angle:

$360° \div 8$
$= 45°$

Mark 8 equal points around the circle and join them.

8. CONSTRUCTING TRIANGLES USING COMPASSES

Example:

AB = 8 cm
AC = 5 cm
BC = 7 cm

Steps:

1. Draw AB = 8 cm.
2. With centre A, draw an arc of radius 5 cm.
3. With centre B, draw an arc of radius 7 cm.
4. Mark their intersection as C.
5. Join A to C and B to C.

9. NETS

A net is a flat 2-dimensional shape that can be folded
to form a 3-dimensional solid.

10. NET OF A CUBE

A cube has:

6 faces

All its faces are equal squares.

A cube net contains:

6 equal squares arranged so that they can fold
to form a cube.

11. NET OF A CUBOID

A cuboid has:

6 rectangular faces.

Its net consists of 6 rectangles arranged so that they can fold to make a cuboid.

12. PRISMS

A prism is a 3-dimensional solid with two identical polygon-shaped ends.

Examples:

- Triangular prism
- Pentagonal prism
- Hexagonal prism

A prism net contains:

- Two identical polygonal ends
- Rectangular faces joining the two ends

13. IMPORTANCE OF ACCURACY

Accurate construction requires:

- Careful measurement
- Proper use of a ruler
- Correct use of a protractor
- Properly adjusted compasses
- A sharp pencil

QUICK REVISION

IMPORTANT SHAPE PROPERTIES

Square:
4 equal sides + 4 right angles

Rectangle:
Opposite sides equal + 4 right angles

Regular polygon:
All sides equal + all angles equal

CENTRE ANGLE:

Centre angle
= 360° ÷ Number of sides

Centre angle = Exterior angle

COMMON CENTRE ANGLES:

Pentagon → 72°
Hexagon → 60°
Heptagon → about 51°
Octagon → 45°
Nonagon → 40°
Decagon → 36°

NET:

A flat 2D shape that folds to form a 3D solid.

Cube net:
6 equal squares

Cuboid net:
6 rectangles

Prism net:
2 identical polygonal ends + rectangular faces

UNIT 13: AREA, SURFACE AREA AND VOLUME

1. AREA OF A CIRCLE

Formula:

Area = πr^2

Where:

r = radius
$\pi \approx 3.14$ or 22/7

Example:

Radius = 5 cm

Area:

$= 3.14 \times 5 \times 5$
$= 78.5\ cm^2$

2. IMPORTANT CIRCLE FACTS

Diameter = 2 × Radius

Radius = Diameter ÷ 2

When diameter is given, first find the radius before using:

$A = \pi r^2$

3. AREA, SURFACE AREA AND VOLUME

Area:
The amount of flat surface enclosed by a boundary.

Surface Area:
The total area covering the outside of a solid.

Volume:
The amount of space occupied by a solid.

4. SURFACE AREA OF A CYLINDER

Closed cylinder:

Surface Area
$= 2\pi r^2 + 2\pi rh$

Open cylinder with one circular end:

Surface Area
$= \pi r^2 + 2\pi rh$

Where:

r = radius
h = height

5. SURFACE AREA OF A PRISM

Formula:

Surface Area
= 2 × Area of base + Area of lateral faces

The two bases are identical.

6. SURFACE AREA OF A PYRAMID

Formula:

Surface Area
= Area of base + Area of triangular/slant faces

7. SURFACE AREA OF A CONE

Curved Surface Area:

$= \pi r l$

Where:

r = radius
l = slant height

Total Surface Area of a closed cone:

$= \pi r l + \pi r^2$

8. SURFACE AREA OF A SPHERE

Formula:

Surface Area $= 4\pi r^2$

9. SURFACE AREA OF A HEMISPHERE

Total surface area:

$= 3\pi r^2$

10. VOLUME OF A PRISM

Formula:

Volume = Base Area × Height

$V = Bh$

Where:

B = area of base
h = height/length of prism

11. VOLUME OF A CYLINDER

Formula:

$V = \pi r^2 h$

Example:

r = 3 cm
h = 5 cm

V:

$= 3.14 \times 3 \times 3 \times 5$
$= 141.3 \text{ cm}^3$

12. VOLUME OF A CONE

A cone has one-third the volume of a cylinder with the same base and height.

Formula:

$V = 1/3\ \pi r^2 h$

Example:

$r = 3$ cm
$h = 4$ cm

V:

$= 1/3 \times 3.14 \times 3 \times 3 \times 4$

$= 37.68$ cm^3

13. VOLUME OF A PYRAMID

A pyramid has one-third the volume of a prism with the same base and height.

Formula:

$V = 1/3 \times \text{Base Area} \times \text{Height}$

$V = 1/3\ Bh$

Example:

Base = 7 cm × 5 cm
Height = 6 cm

Base area:

$7 \times 5 = 35$ cm^2

Volume:

$= 1/3 \times 35 \times 6$

$= 70$ cm^3

14. VOLUME OF A SPHERE

Formula:

$V = 4/3\ \pi r^3$

Example:

$r = 3$ cm

V:

$= 4/3 \times 3.14 \times 3 \times 3 \times 3$

$= 113.04\ cm^3$

15. VOLUME OF A HEMISPHERE

A hemisphere is half a sphere.

Formula:

$V = 2/3\ \pi r^3$

16. CORRECT UNITS

AREA uses SQUARE units:

mm^2
cm^2
m^2
km^2

VOLUME uses CUBIC units:

mm^3
cm^3
dm^3
m^3

REMEMBER:

Area → square units
Volume → cubic units

QUICK REVISION

CIRCLE:

$A = \pi r^2$

CYLINDER:

Surface Area:
$= 2\pi r^2 + 2\pi rh$

Volume:
$= \pi r^2 h$

CONE:

Curved Surface Area:
$= \pi rl$

Total Surface Area:
$= \pi rl + \pi r^2$

Volume:
$= 1/3\ \pi r^2 h$

PRISM:

Volume:
$= Bh$

PYRAMID:

Volume:
$= 1/3\ Bh$

SPHERE:

Surface Area:
$= 4\pi r^2$

Volume:
$= 4/3\ \pi r^3$

HEMISPHERE:

Total Surface Area:
$= 3\pi r^2$

Volume:

$= 2/3 \pi r^3$

VERY IMPORTANT:

Cone volume
= 1/3 of a cylinder with the same base and height.

Pyramid volume
= 1/3 of a prism with the same base and height.

UNIT 14: DATA HANDLING

1. DATA

Data is information collected for a particular purpose.

Data may include:

- Numbers
- Words
- Names
- Colours
- Pictures

Data helps people understand situations and make decisions.

2. DATA COLLECTION

Data collection is the organised process of gathering information to answer a question.

Example:

A teacher may collect learners' marks to find:

- How many learners passed
- The most common mark
- The total number of learners

3. TALLY MARKS

A tally is a way of counting using strokes.

Example:

| | | | = 4

Tallies are usually grouped in fives.

Tally marks make counting easier.

4. FREQUENCY

Frequency is the number of times a value or item occurs.

Example:

If 40 kg appears 9 times:

Frequency = 9

5. FREQUENCY TABLE

A frequency table shows:

• Values or categories
• Their frequencies

Example:

Mass (kg)	Frequency
28	4
30	7
33	5
35	6
40	9
42	1

Total = 32

The most common mass is:

40 kg

6. BAR CHART

A bar chart represents data using bars.

Bars may be:

- Vertical
- Horizontal

The height or length of each bar represents the quantity.

7. PARTS OF A GOOD BAR CHART

A good bar chart should have:

- A title
- Horizontal axis
- Vertical axis
- Suitable scale
- Bars of equal width
- Equal spaces between bars

8. INTERPRETING A BAR CHART

A bar chart can help us find:

- Highest value
- Lowest value
- Total
- Difference between categories
- Most common category

9. PIE CHART

A pie chart is a circle divided into sectors.

The whole circle represents:

$360° = 100\% = 1$ whole

10. FINDING A SECTOR ANGLE

Formula:

Sector angle
= Frequency / Total frequency × 360°

Example:

15 learners out of 60 choose Mathematics.

Sector angle:

$15/60 \times 360°$

$= 90°$

11. FINDING PERCENTAGE IN A PIE CHART

Formula:

Percentage
$= \text{Frequency} / \text{Total frequency} \times 100\%$

Example:

15 out of 60:

$15/60 \times 100\%$

$= 25\%$

12. FINDING FREQUENCY FROM A SECTOR

Formula:

Frequency
$= \text{Sector angle} / 360° \times \text{Total frequency}$

Example:

Sector angle = 90°
Total = 40

Frequency:

$90/360 \times 40$

$= 10$

13. INTERPRETING PIE CHARTS

To interpret a sector:

Number represented
= Sector fraction × Total

Percentage
= Sector fraction × 100%

Example:

Sector angle = 216°

Fraction:

216/360
= 3/5

Percentage:

3/5 × 100%
= 60%

14. REPRESENTING DATA IN DIFFERENT WAYS

The same data can be shown using:

- Frequency table
- Bar chart
- Pie chart

Recommended steps:

1. Collect the data.
2. Categorise the data.
3. Count each occurrence.
4. Make a frequency table.
5. Use a bar chart to compare categories.
6. Use a pie chart to show parts of a whole.

QUICK REVISION

Data
= Information collected for a purpose.

Tally
= Strokes used for counting.

Frequency
= Number of times something occurs.

Frequency table
= Table showing values/categories and frequencies.

Bar chart
= Graph using bars to represent quantities.

Pie chart
= Circle divided into sectors.

Scale
= Value represented by each division on an axis.

IMPORTANT PIE CHART FORMULAS

Sector angle
= Frequency / Total frequency × 360°

Percentage
= Frequency / Total frequency × 100%

Frequency
= Sector angle / 360° × Total frequency

REMEMBER:

Whole pie chart:

360° = 100% = 1 whole

UNIT 15: LIKELIHOOD OF EVENTS

1. PROBABILITY

Probability describes how likely an event is to happen.

Different events have different chances of occurring.

Example:

• The sun rising tomorrow → Certain
• Getting a head when tossing a fair coin → Equally likely
• Rolling a 7 on a normal six-sided die → Impossible

2. RANDOM EVENT

A random event is an event whose outcome cannot always be predicted with certainty.

Examples:

• Tossing a coin
• Rolling a die
• Picking an object without looking

3. IMPOSSIBLE EVENT

An impossible event cannot happen.

Chance:

0 or 0%

Examples:

• Rolling a 7 on a six-sided die.
• Getting both a head and tail from one toss of one coin.

4. CERTAIN EVENT

A certain event will definitely happen.

Chance:

1 or 100%

Example:

The sun will rise in the morning.

5. LIKELY EVENT

A likely event has a good chance of happening.

It is more likely to happen than not happen.

Example:

A bag contains:

8 red balls
2 blue balls

Picking a red ball is LIKELY because there are more red balls.

6. UNLIKELY EVENT

An unlikely event has a small chance of happening.

It can happen, but it is not expected to happen often.

Using the same bag:

8 red balls
2 blue balls

Picking a blue ball is UNLIKELY.

7. EQUALLY LIKELY EVENTS

Events are equally likely when they have the same chance of happening.

Example:

A fair coin has:

Head
Tail

Getting a head and getting a tail are equally likely.

8. ORDER OF LIKELIHOOD

Events can be arranged from least likely to most likely:

IMPOSSIBLE
↓
UNLIKELY
↓
EQUALLY LIKELY
↓
LIKELY
↓
CERTAIN

9. PROBABILITY SCALE

0 ---------------- 1/2 ---------------- 1

Impossible Equal chance Certain

0% -------------- 50% -------------- 100%

10. COMPARING EVENTS

Example:

A bag contains:

6 green balls
3 yellow balls
1 red ball

Without looking:

Green → Most likely
Yellow → Less likely than green
Red → Least likely

The more objects of one type there are, the greater the chance of selecting that type.

11. PROBABILITY IN REAL LIFE

Probability can be used when thinking about:

• Weather
• Games
• Sports results
• Tossing coins
• Rolling dice
• Selecting objects randomly

However, random events cannot always be predicted with certainty.

QUICK REVISION

Probability
= How likely an event is to happen.

Impossible
= Cannot happen
= 0 or 0%

Unlikely
= Small chance of happening

Equally likely
= Same chance of happening

Likely
= Good chance of happening

Certain
= Definitely happens
= 1 or 100%

ORDER:

Impossible → Unlikely → Equally likely → Likely → Certain

PROBABILITY SCALE:

0 → Impossible

1/2 → Equal chance

1 → Certain

REMEMBER:

Impossible = 0%

Certain = 100%

Random events cannot always be predicted with certainty.